

A Practical Guide to Cross-validation statistics

TOPIC -- CROSS-VALIDATION STATISTICS

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```
err ← 0
for i ← 1, ..., N do // define the cross-validation subsets
  x_in ← (x[1], ..., x[i - 1], x[i + 1], ..., x[N])
  y_in ← (y[1], ..., y[i - 1], y[i + 1], ..., y[N])
  x_out ← x[i]
  y_out ← interpolate(x_in, y_in, x_out)
  err ← err + (y[i] - y_out)^2
end for
err ← err/N
```

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$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{a} - \boldsymbol{\beta}^T \mathbf{x}_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{a} - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2$$

$$\begin{aligned} \text{MSE} &= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{a} - \boldsymbol{\beta}^T \mathbf{x}_i)^2 \end{aligned}$$

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Motivation Assume a model with one or more unknown parameters, and a data set to which the model can be fit (the training data set). The fitting process optimizes the model parameters to make the model fit the training data as well as possible. If an independent sample of validation data is taken from the same population as the training data, it will generally turn out that the model does not fit the validation data as well as it fits the training data. The size of this difference is likely to be large especially when the size of the training data set is small, or when the number of parameters in the model is large. Cross-validation is a way to estimate the size of this effect.

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Relative accuracy can be quantified as $\text{MSE}(\lambda_i) / \text{MSE}(\lambda_R)$, so that the mean squared error of a candidate λ_i is made relative to that of a user-specified λ_R . The relative simplicity term measures the amount that λ_i deviates from λ_R relative to the maximum amount of deviation from λ_R . Accordingly, relative simplicity can be specified as $\frac{(\lambda_i - \lambda_R)^2}{(\lambda_{\max} - \lambda_R)^2}$, where $\lambda_{\max}$

$\lambda_{\max}$ corresponds to the λ value with the highest permissible deviation from λ_R . With $\gamma \in [0, 1]$, the user determines how high the influence of the reference parameter is relative to cross-validation. One can add relative simplicity terms for multiple configurations $c = 1, 2, \dots, C$ by specifying the loss function as

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Example: linear regression In linear regression, there exist real response values $y_1, \dots, y_n$, and n p -dimensional vector covariates $x_1, \dots, x_n$. The components of the vector x_i are denoted $x_{i1}, \dots, x_{ip}$. If least squares is used to fit a function in the form of a hyperplane $\hat{y} = a + \beta^T x$ to the data (x_i, y_i) $1 \leq i \leq n$, then the fit can be assessed using the mean squared error (MSE). The MSE for given estimated parameter values a and β on the training set (x_i, y_i) $1 \leq i \leq n$ is defined as:

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Using prior information When users apply cross-validation to select a good configuration λ , then they might want to balance the cross-validated choice with their own estimate of the configuration. In this way, they can attempt to counter the volatility of cross-validation when the sample size is small and include relevant information from previous research. In a forecasting combination exercise, for instance, cross-validation can be applied to estimate the weights that are assigned to each forecast. Since a simple equal-weighted forecast is difficult to beat, a penalty can be added for deviating from equal weights. Or, if cross-validation is applied to assign individual weights to observations, then one can penalize deviations from equal weights to avoid wasting potentially relevant information. Hoornweg (2018) shows how a tuning parameter γ can be defined so that a user can intuitively balance between the accuracy of cross-validation and the simplicity of sticking to a reference parameter λ_R that is defined by the user. If λ_i denotes the i th candidate configuration that might be selected, then the loss function that is to be minimized can be defined as

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If the model is correctly specified, it can be shown under mild assumptions that the expected value of the MSE for the training set is $(n - p - 1)/(n + p + 1) < 1$ times the expected value of the MSE for the validation set (the expected value is taken over the distribution of training sets). Thus, a fitted model and computed

MSE on the training set will result in an optimistically biased assessment of how well the model will fit an independent data set. This biased estimate is called the in-sample estimate of the fit, whereas the cross-validation estimate is an out-of-sample estimate. Since in linear regression it is possible to directly compute the factor $(n - p - 1)/(n + p + 1)$ by which the training MSE underestimates the validation MSE under the assumption that the model specification is valid, cross-validation can be used for checking whether the model has been overfitted, in which case the MSE in the validation set will substantially exceed its anticipated value. (Cross-validation in the context of linear regression is also useful in that it can be used to select an optimally regularized cost function.)

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Cross-validation, sometimes called rotation estimation or out-of-sample testing, is any of various similar model validation techniques for assessing how the results of a statistical analysis will generalize to an independent data set. Cross-validation includes resampling and sample splitting methods that use different portions of the data to test and train a model on different iterations. It is often used in settings where the goal is prediction, and one wants to estimate how accurately a predictive model will perform in practice. It can also be used to assess the quality of a fitted model and the stability of its parameters. In a prediction problem, a model is usually given a dataset of known data on which training is run (training dataset), and a dataset of unknown data (or first seen data) against which the model is tested (called the validation dataset or testing set). The goal of cross-validation is to test the model's ability to predict new data that was not used in estimating it, in order to flag problems like overfitting or selection bias and to give an insight on how the model will generalize to an independent dataset (i.e., an unknown dataset, for instance from a real problem). One round of cross-validation involves partitioning a sample of data into complementary subsets, performing the analysis on one subset (called the training set), and validating the analysis on the other subset (called the validation set or testing set). To reduce variability, in most methods multiple rounds of cross-validation are performed using different partitions, and the validation results are combined (e.g. averaged) over the rounds to give an estimate of the model's predictive performance. In summary, cross-validation combines (averages) measures of fitness in prediction to derive a more accurate estimate of model prediction performance.